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$$\begin{aligned} & \int (x^2 + x - 2)e^x dx \\ & \begin{cases} u = (x^2 + x - 2) \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = (2x + 1)dx \\ v = e^x \end{cases} \\ & \Rightarrow (x^2 + x - 2)e^x - \int (2x + 1)e^x dx \\ & \begin{cases} u = 2x + 1 \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = 2dx \\ v = e^x \end{cases} \\ & \Rightarrow (x^2 + x - 2)e^x - \left((2x + 1)e^x - 2 \int e^x dx \right) \\ & = x^2 e^x + x e^x - 2e^x - 2x e^x - e^x + 2e^x + C \\ & = (x^2 - x - 1)e^x + C \end{aligned}$$

References